Year 12 Further Maths - Complex numbers

Similar questions - Arithmetic, conjugates and polar form

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning. The complex numbers z_1 and z_2 are given by $z_1 = 4 - 3\mathrm{i}$ and $z_2 = a + 2\mathrm{i}$, where a is a real number.

(a) Express z_1 in modulus-argument form, giving the modulus exactly and the argument to 3 significant figures. **[3]**

(b) Find $z_1 z_2$ in terms of a, in the form $c + \mathrm{i}d$. **[2]**

(c) The real and imaginary parts of a complex number on an Argand diagram are x and y. Given that $z_1 z_2$ lies on the line $y = x$, find a. **[2]**

(d) Given instead that $z_1 z_2 = (z_1 z_2)^*$, find a. **[2]**

2 You are given that $z = 5 - 12\mathrm{i}$.

(a) Find $|z|$, $\arg z$ and z^*. **[3]**

(b) On an Argand diagram w is represented by A and w^* by B. Describe the geometrical relationship between A and B. **[2]**

3 In this question the principal argument of a complex number lies in $[0, 2\pi)$. Complex numbers $z_1 = 5 + 12\mathrm{i}$ and $z_2 = -4 + 3\mathrm{i}$.

(a) Determine $z_1 z_2$ in the form $a + b\mathrm{i}$. **[2]**

(b) Express z_2 in modulus-argument form. **[3]**

(c) Verify by direct calculation that $\arg(z_1 z_2) = \arg z_1 + \arg z_2$. **[3]**

4 In this question you must show detailed reasoning.

(a) Express $\dfrac{8 + \mathrm{i}}{2 - \mathrm{i}}$ in the form $a + b\mathrm{i}$ where a and b are real. **[2]**

(b) Solve the equation $4x^2 - 8x + 5 = 0$, giving your answers in the form $c + d\mathrm{i}$. **[2]**

(c) Solve $2w^2 - 4w + 5 = 0$. **[2]**

Question 1
(a) $|z_1| = 5$, $\arg z_1 = -\tan^{-1}\frac{3}{4} = -0.644$. $z_1 = 5\left(\cos(-0.644) + \mathrm{i}\sin(-0.644)\right)$
(b) $(4-3\mathrm{i})(a+2\mathrm{i}) = (4a+6)+(8-3a)\mathrm{i}$
(c) $4a+6 = 8-3a \Rightarrow a = \frac{2}{7}$
(d) $z_1 z_2$ is real when $8-3a = 0$, $a = \frac{8}{3}$

Question 2
(a) $|z| = 13$, $\arg z = -\tan^{-1}\frac{12}{5} = -1.18$ (rad), $z^* = 5+12\mathrm{i}$
(b) B is the reflection of A in the real axis

Question 3
(a) $z_1 z_2 = -20+15\mathrm{i}-48\mathrm{i}-36 = -56-33\mathrm{i}$
(b) $|z_2| = 5$, second quadrant: $\arg z_2 = \pi - \tan^{-1}\frac{3}{4} = 2.50$. $z_2 = 5(\cos 2.50 + \mathrm{i}\sin 2.50)$
(c) $\arg z_1 = \tan^{-1}\frac{12}{5} = 1.176$, so $\arg z_1 + \arg z_2 = 3.674$. $z_1 z_2 = -56-33\mathrm{i}$ is in the third quadrant: $\arg = \pi + \tan^{-1}\frac{33}{56} = 3.674$ (equal)

Question 4
(a) $\dfrac{(8+\mathrm{i})(2+\mathrm{i})}{5} = \dfrac{16+8\mathrm{i}+2\mathrm{i}-1}{5} = 3+2\mathrm{i}$
(b) $x = \dfrac{8 \pm \sqrt{64-80}}{8} = 1 \pm \frac{1}{2}\mathrm{i}$
(c) $w = \dfrac{4 \pm \sqrt{16-40}}{4} = 1 \pm \dfrac{\sqrt{6}}{2}\mathrm{i}$